

**THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS**

MSBA7027 Machine Learning

GENERAL INFORMATION	
Instructor: Zhengli Wang Email: wzl1@hku.hk Office: KK908 Phone: 3917-1034 Consultation times: TBA Tutor: TBA Pre-requisites: MSBA7002 Course Website: Moodle The following programming languages are used in this course: R.	
COURSE DESCRIPTION	
This course provides systematic training of machine learning, which is the foundation and core of data analytics. It covers various topics such as machine learning system design, nonlinear regressions, advanced tree-based methods, recommendation systems and anomaly detection.	
COURSE OBJECTIVES	
1. Obtain detailed understanding of machine learning. 2. Learn algorithms. 3. Implement algorithms in R. 4. Obtain understanding of the algorithms used in real life.	
PROGRAMME LEARNING OUTCOMES	
PLO1: Acquisition and internalization of knowledge of the programme discipline PLO2: Application and integration of knowledge PLO3: Inculcating professionalism and leadership PLO4: Developing global outlook PLO5: Mastering communication skills	
COURSE LEARNING OUTCOMES	
Course Learning Outcomes	Aligned Programme Learning Outcomes

CLO1: Clearly identify and formulate the relevant problem	PLO1
CLO2: Select and use effective methods to address the problem	PLO2
CLO3: Use software tools to provide solution to the issue at hand	PLO2, 4
CLO4: Communicate the solution effectively	PLO3, 5

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities	Expected contact hour	Study Load (% of study)
T&L1. Interactive lectures	30	25%
T&L2. Assignments, tutorials	60	50%
T&L 3 Self-study, group discussion	30	25%
Total	120	100%

Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes
A1. Participation	Attendance & discussions	20%	CLO1, 2, 3, 4
A2. Assignments	Effort and accuracy	40%	CLO1, 2, 3, 4
A3. Final exam	Effort and accuracy	40%	CLO1, 2, 3, 4
	Total	100%	

STANDARDS FOR ASSESSMENT

Course Grade Descriptors

A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions • Present arguments that have an element of originality • Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills • Excellent performance in assignments
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Good participation in discussions • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills • Good performance in assignments
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Some participation in discussions • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills • Acceptable performance in assignments

D+, D	• Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Minimal or no participation in discussions • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills • Marginally acceptable performance in assignments
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F	• Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Minimal or no participation in discussions • Present arguments poorly • Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills • Poorly performance in assignments
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Assessment Rubrics for Participation

A+, A, A-	• High participation in discussions • Always attend in-class discussions • Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • Present arguments that have an element of originality • Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	• Good participation in discussions • Often attend in-class discussions • Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Present arguments that go beyond the lecture and textbook • Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	• Some participation in discussions • Sometimes attend in-class discussions • Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Present arguments in a well-structure manner • Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	• Minimal or no participation in discussions • Rarely attend in-class discussions • Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Present arguments in a marginally acceptable manner • Respect others and follow the class rules (no chatting and do not use cell phone)
F	• Minimal or no participation in discussions • Almost never attend in-class discussions • Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Present arguments poorly • Behave poorly in class (often chatting with others, using cell phones, or being late)

Assessment Rubrics for Assignments and the Exam

A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
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B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills
D+, D	• Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	• Demonstrate a poor understanding of the concepts involved • Present arguments poorly • Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE	
Lecture 1 Machine Learning Overview • Machine Learning Trend / Applications • Recap of Core Concepts: ○ Supervised vs Unsupervised / Parametric vs Nonparametric ○ Model Performance Assessment (Confusion Matrix, Precision/Recall, F-score, Accuracy, AUC) ○ Model Validation and Selection / Bias-Variance ○ Other Concepts Lecture 2 Review of Fundamental Algorithms • Linear Regression / Logistic Regression / Clustering / SVM (support vector machine) • Pros and cons of each method / Limitations of linear models Lecture 3-4 Nonlinear Methods • Nonlinear Methods in Regression: basis expansion and kernel methods • Nonlinear Methods in SVM Lecture 5 Machine Learning System Design • Data Preprocessing • Feature Engineering (One-Hot Encoding, Binning, Normalization, Standardization, Missing Values, Imputation) • Feature Selection • Resampling Methods, Hyperparameter Tuning Lecture 6 Tree, Bagging, Random Forest • Decision Trees, Splitting, Pruning, Gini Impurity • Bagging, Random Forest Lecture 7 Gradient Boosting Machines • Gradient Boosting Machine • Stochastic Gradient Boosting Machine • XGBoost Lecture 8 Ensemble Learning/Stacking • Super Learner Algorithm Lecture 9 Interpretable Machine Learning • Categorization: Agnosticity, Scope, Explanation Type • Permutation-based Feature Importance • Partial Dependence Plot • Individual conditional expectation • LIME • SHAP Lecture 10 Recommendation System • Recommendation Systems • Collaborative Filtering & Algorithms • Low Rank Matrix Factorization • Real Life Applications	

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

Andriy Burkov. The Hundred-Page Machine Learning Book. Hard Cover ed. edition (January 11, 2019).

Christoph Molnar. Interpretable Machine Learning. <https://christophm.github.io/interpretable-ml-book/>

Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, *An Introduction to Statistical Learning with Application in R*, First Edition, 2013, Springer. <http://www-bcf.usc.edu/~gareth/ISL/>.

Trevar Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Second Edition, 2008, Springer.
<http://statweb.stanford.edu/~tibs/ElemStatLearn/download.html>.

Christopher Bishop. Pattern Recognition and Machine Learning, Springer.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

☒ conducting mid-term survey in addition to SETL around the end of the semester

☐ Online response via Moodle site

☐ Others: _____ (please specify)

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

Plagiarism will not be tolerated.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

Late homework will not be accepted.